

Optima - User Manual

Smart Merchandising Platform - Complete Guide

DOCUMENT TYPE

Complete Operations Manual

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v1.0 (Official)

TARGET SYSTEM

Predictive Sales & Bundling Platform

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1. System Overview

Optima is a smart merchandising platform that converts raw sales transaction data into strategic retail intelligence. It provides three core capabilities:

Capability	Description
Sales Forecasting	Trains a deep learning model (DeepAR) on your historical data to predict future product demand over a 12-month horizon.
Product Bundler	Uses association rule mining (Apriori algorithm) to identify which products are frequently bought together, then categorizes those pairings as strategic bundles.
Analytics Dashboard	Presents interactive charts, year-over-year benchmarking, model health scores, and STL trend decomposition for each forecast run.

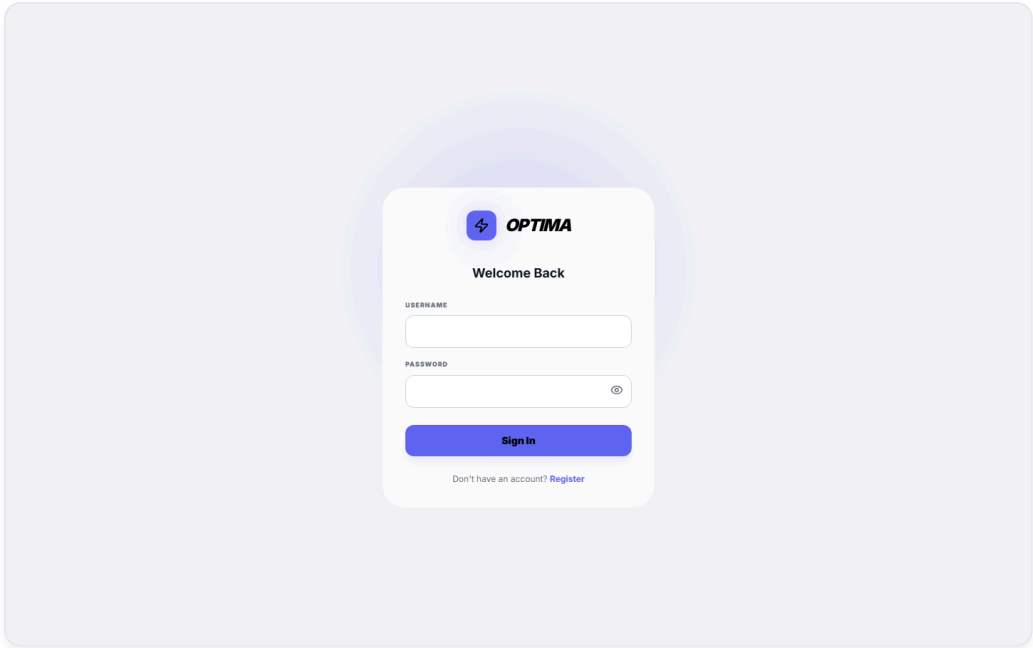
The system uses a **role-based access model**:

- **USER** - Can upload data, train models, view analytics, and explore bundles.
 - **ADMIN** - All USER capabilities plus full user management, session control, and system audit trails.
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2. Getting Started - Login & Registration

2.1 Signing In

Navigate to www.metatech.live . You will land on the **Sign In** page.



Field / Button	Description
Username field	Enter your registered username.
Password field	Enter your password.
Eye icon (inside password field)	Toggles password visibility between hidden (dots) and plain text.
Sign In button	Submits your credentials. On success, you are redirected to the Management Hub. On failure, a red error banner appears below the logo.
Register link	Navigates to the Registration page if you do not yet have an account.

NOTE

If your account is **pending approval**, the Sign In button will return an error indicating your account has not been approved yet. Contact your system administrator.

2.2 Registering an Account

Click the **Register** link on the Sign In page. You will see a multi-field registration form.

Field	Description
Username	Your unique login handle. Cannot be changed after registration.
First Name / Last Name	Your display name shown in the sidebar.
Middle Name	Optional.
Email	Used for profile identification.
Phone Number	Optional contact detail.
Password	Must be at least 8 characters, and include: one uppercase letter, one lowercase letter, one number, and one special character (!@#\$%^&* etc.).
Confirm Password	Must match the password field exactly.
Register button	Submits the form. Your account status will be pending until an administrator approves it.

IMPORTANT

New accounts **require administrator approval** before they can log in. After registering, wait for an admin to approve your account through the Admin Panel.

3. Navigation & Sidebar

Once logged in, you will see a fixed left sidebar.

3.1 Sidebar Structure

OPTIMA	← Logo / app name
ANALYTICAL PIPELINE	
Management Hub	← Data & model management
Forecasting	← View & open forecast runs
Product Bundler	← View & explore bundle strategies
SYSTEM (Admin only)	
Admin Panel	← User management, audit logs

3.2 Sidebar Elements

Element	Description
OPTIMA logo (top)	Decorative; always visible when logged in.
Active nav link (indigo/accent background)	Indicates the current page.
Inactive nav link	Clicking navigates to that section. Hover shows a subtle background.
Account card (bottom of sidebar)	Displays your name and role. Click to open the profile menu.
Up / Down chevron (on account card)	Indicates whether the profile menu is open or closed.

3.3 Profile Menu (Account Card)

Click the **account card** at the bottom of the sidebar to expand the profile menu.

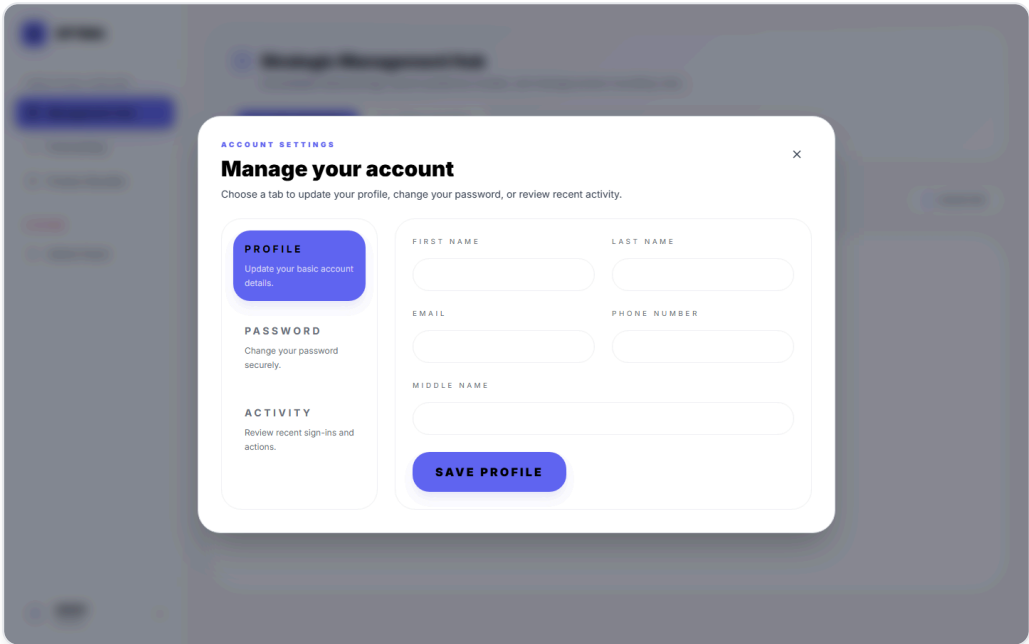
Button	Description
Account Settings	Opens the Account Settings modal (see Section 4).
Light Mode / Dark Mode	Toggles between light and dark interface themes. The current opposite is shown as the option.
View as User <i>(Admin only)</i>	Allows an administrator to temporarily see the interface as a regular USER would (hides the Admin Panel link). Active when checked (amber checkmark appears).
Logout (red)	Ends your session and returns you to the Login page.

4. Account Settings

Access via: **Account Card** → **Account Settings**

A full-screen modal opens with three tabs on the left side.

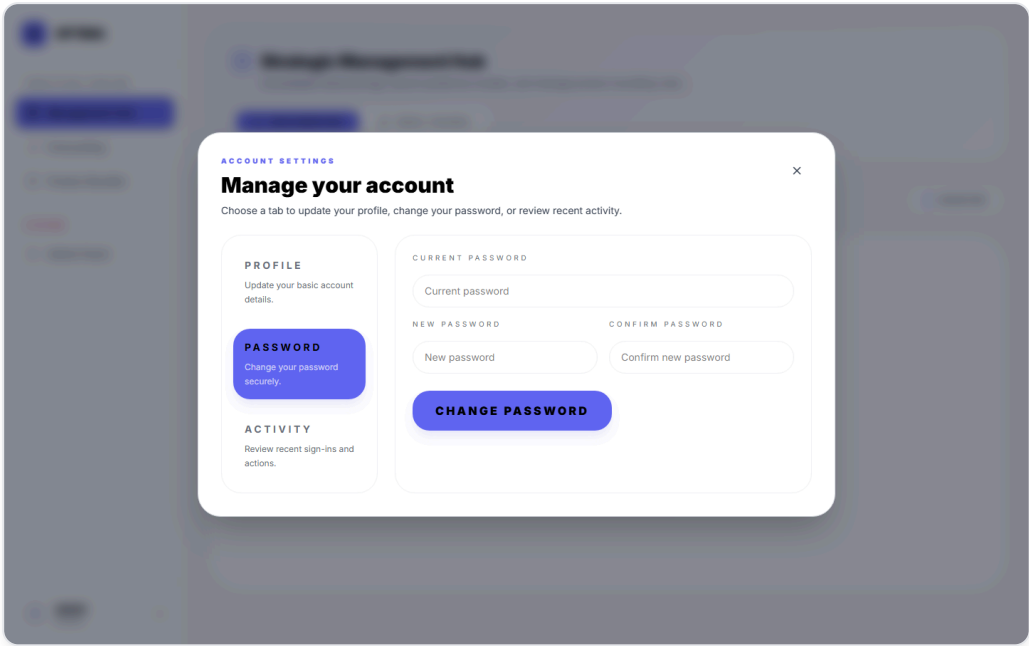
4.1 Profile Tab



Update your personal information.

Field	Description
First Name	Editable. Shown in the sidebar next to your username.
Last Name	Editable.
Email	Editable.
Phone Number	Editable.
Middle Name	Optional, spans full width.
Save Profile button	Saves all profile changes. Shows "Saving..." while the request is in progress. Disabled (grayed out) while loading.

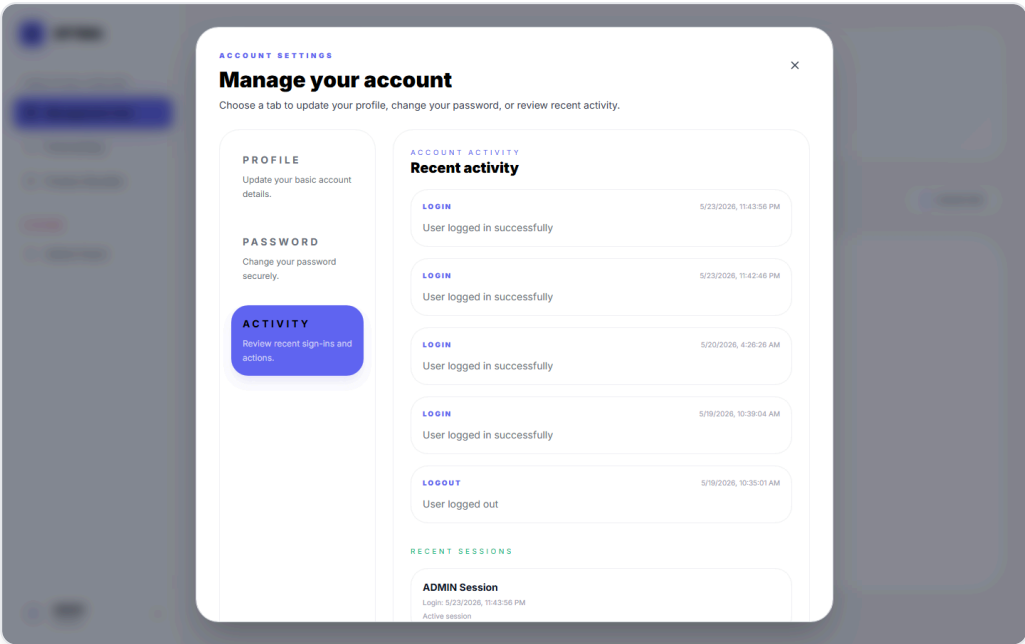
4.2 Password Tab



Change your login password.

Field	Description
Current Password	Enter your existing password to authorize the change.
New Password	Your new password. Must meet the same strength requirements as registration.
Confirm Password	Must exactly match New Password .
Change Password button	Submits the change. Shows "Updating..." while processing. If passwords don't match or are too weak, a red error banner appears. On success, a green banner confirms the update.

4.3 Activity Tab



A read-only log of recent account events (most recent 5 of each).

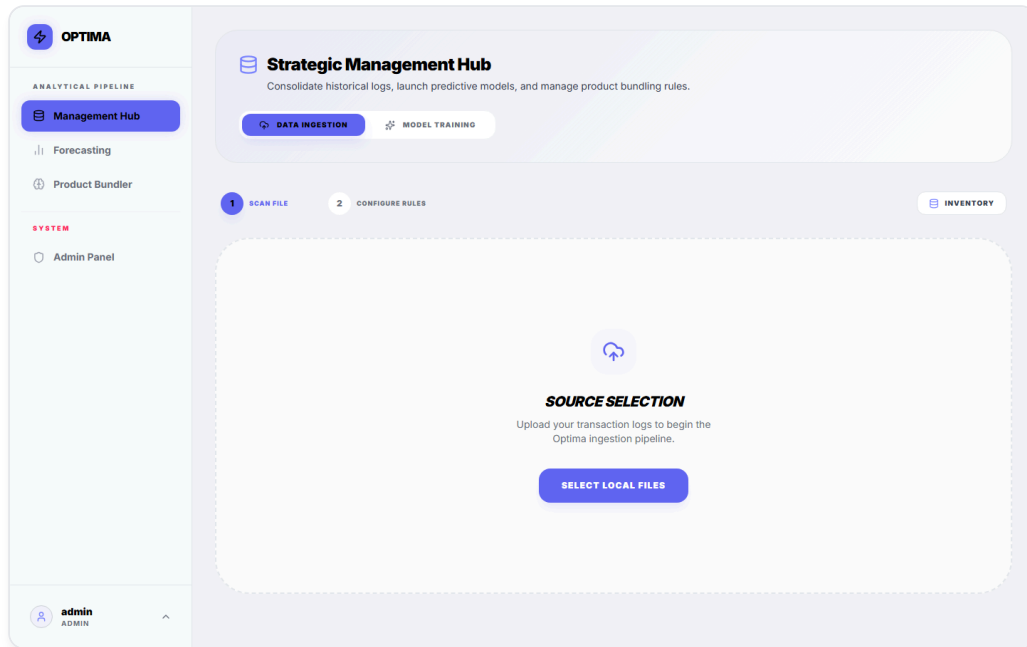
Section	Description
Recent Activity	Lists audit log events (e.g., login, password change) with their action name and timestamp.
Recent Sessions	Lists login sessions showing login time, logout time (or "Active session" if still open), and the role used.

NOTE

To close the modal, click the **X** button in the top-right, or click anywhere outside the modal panel.

5. Management Hub - Data Ingestion

Navigate to **Management Hub** (/) in the sidebar.



The hub has two main **tabs** at the top:

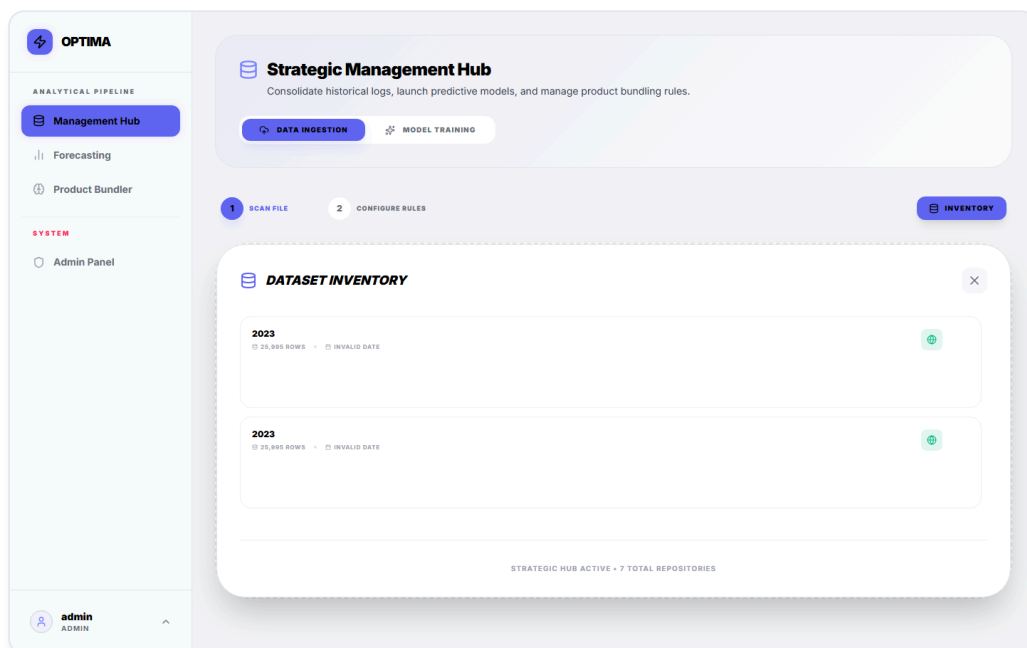
Tab	Description
Data Ingestion	Upload new transaction data files.
Model Training	Train forecasting and bundling models on your datasets.

5.1 Data Ingestion Tab - Step 1: Scan File

The default view shows a large dashed drop zone.

Element	Description
Select Local Files button (indigo)	Opens a file browser. Supports .csv , .xlsx , and .xls files. You can select multiple files at once.
Analyzing Files... (spinner)	Appears while Optima scans the uploaded files to detect all unique product/item names.
Inventory button (top-right, database icon)	Toggles the Dataset Inventory panel, which slides over the main area showing all your existing datasets.

Dataset Inventory Panel



When **Inventory** is active, a panel overlays the drop zone showing all saved datasets.

Element	Description
Dataset card (each row)	Shows the dataset title, row count, upload date, and last-edited date.
Edit icon (appears on hover, accent color)	Click to rename the dataset inline.
Save icon (appears after clicking edit)	Confirms the new name.
Cancel icon (appears after clicking edit)	Discards the rename.
Lock / Globe icon	Toggles the dataset between Private (lock, amber) and Public (globe, green). Private datasets are not visible to other users.
Trash icon (appears on hover, red)	Permanently deletes the dataset and all associated model runs. A confirmation dialog appears first.
Explorer button (indigo)	Opens the Data Viewer for this dataset (see below).
Configure button (muted)	Opens the Item Configuration modal to update product tagging rules for this dataset.
Close button (top-right of panel)	Closes the Inventory panel.

5.2 Data Ingestion Tab - Step 2: Configure Rules

After files are scanned, the UI transitions to the **Ingestion Rules** step.

Element	Description
Step indicator (numbered circles: 1 → 2)	Shows which step of the upload wizard you are on. Completed steps are filled with the accent color.
Back link	Returns to Step 1 (file selection) without losing the scanned file.
Item list	Each detected product/item name appears as a card.

Per-Item Configuration Checkboxes

Each item card has two toggle options:

Checkbox	Description
Bundle (indigo)	Mark this item as a bundle (a pre-packaged product group). When checked, "Exclude" is automatically unchecked. Bundled items are treated as compound products in analysis.
Exclude (red)	Remove this item from all analysis. Use for non-product line items (e.g., delivery fees, discounts). When checked, "Bundle" is automatically unchecked.

NOTE

Leave both unchecked for ordinary individual products. Only mark special items.

Finalizing the Upload

Element	Description
Dataset Display Title input	A required name for this dataset (e.g., "Sales Q1 2026"). Shown everywhere the dataset is referenced.
Commit Dataset button (indigo, full-width)	Uploads and processes the files with your configuration. A progress bar appears during upload. The dataset is automatically activated (set as the active dataset) after a successful commit.
Cancel Upload link	Cancels an in-progress upload.

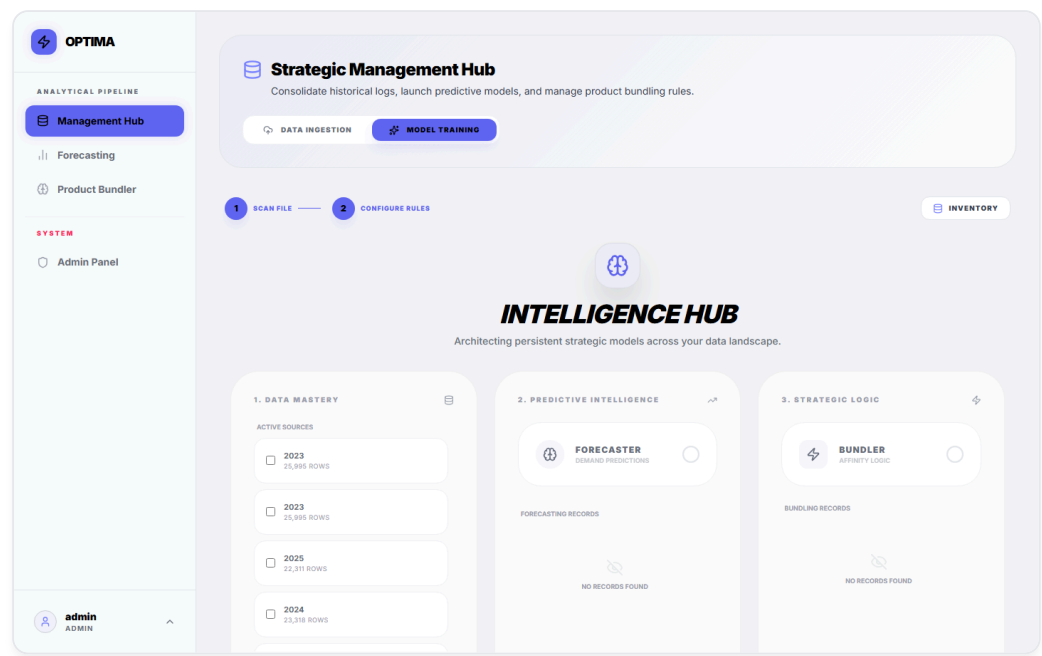
5.3 Data Viewer

Opened from the Inventory panel → **Explorer** button. The viewer displays raw, aggregated, or globally aggregated dataset data in a paginated table.

Control	Description
View toggle (Raw / Aggregated / Global Aggregated)	Switches between raw transaction rows, per-item aggregated summaries, or globally aggregated data.
Year filter dropdown (<i>Raw view only</i>)	Filters rows by a specific transaction year.
Column headers (clickable)	Click any column header to sort by that column. Clicking again toggles between ascending and descending order.
Left / Right Pagination arrows	Navigate between pages (50 rows per page).
Page jump input + Enter	Type a specific page number and press Enter to jump directly to it.
Close button	Closes the viewer.

6. Management Hub - Model Training

Click the **Model Training** tab in the Management Hub header.



The Intelligence Hub is divided into **three columns**.

Column 1 - Data Mastery (Dataset Selection)

Element	Description
Dataset list (checkboxes)	Shows all MASTER datasets. Check one or more to include them in the training run. Selected datasets are highlighted with an accent border.
Configure Selections button	Opens the Item Configuration modal for the first selected dataset. Only enabled when at least one dataset is selected.

Column 2 - Predictive Intelligence (Forecasting)

Element	Description
Forecaster toggle card	Click this card to enable or disable the Sales Forecaster (DeepAR). When enabled, the card glows with an accent border.
Forecasting Records list	Displays a scrollable history of completed forecast runs. Shows each run's name, creation timestamp, and completion status. Features inline rename text fields and delete (trash) buttons.

Column 3 - Product Bundler (Association Mining)

Element	Description
Product Bundler toggle card	Click to enable or disable the Apriori Bundler . When enabled, it glows.
Min. Support (%) slider (<i>appears when bundler is enabled</i>)	Sets the minimum support threshold. Lower values discover more (potentially weaker) bundles; higher values find only very frequent pairings.
Bundler Persistence / Save Bundler Run button (<i>appears when bundler is enabled and forecasting is disabled</i>)	Toggles whether to save the individual bundler run. If forecasting is enabled, bundler runs are automatically saved with the forecast.
Auto-save toggle (<i>appears when bundler is enabled</i>)	When ON, results are immediately committed to the Strategy Vault. When OFF, results appear in the Sandbox for review before saving.
Bundling Records list	Displays a scrollable history of completed bundler runs. Shows each run's name, creation timestamp, and session status. Features inline rename text fields and delete (trash) buttons.

Run Configuration & Launch

Element	Description
Run Name input	A required label for this training session (e.g., "Full Year 2025 Forecast").
Launch Training button (full-width, indigo)	Starts model training. Shows a spinner and progress bar during training.
Cancel link	Cancels an in-progress training.

IMPORTANT

Training can take **up to 30 minutes** for large datasets. Do not close the browser during training. If the connection drops mid-run, the backend may still be processing - check the [Forecasting](#) or [Product Bundler](#) pages before retrying.

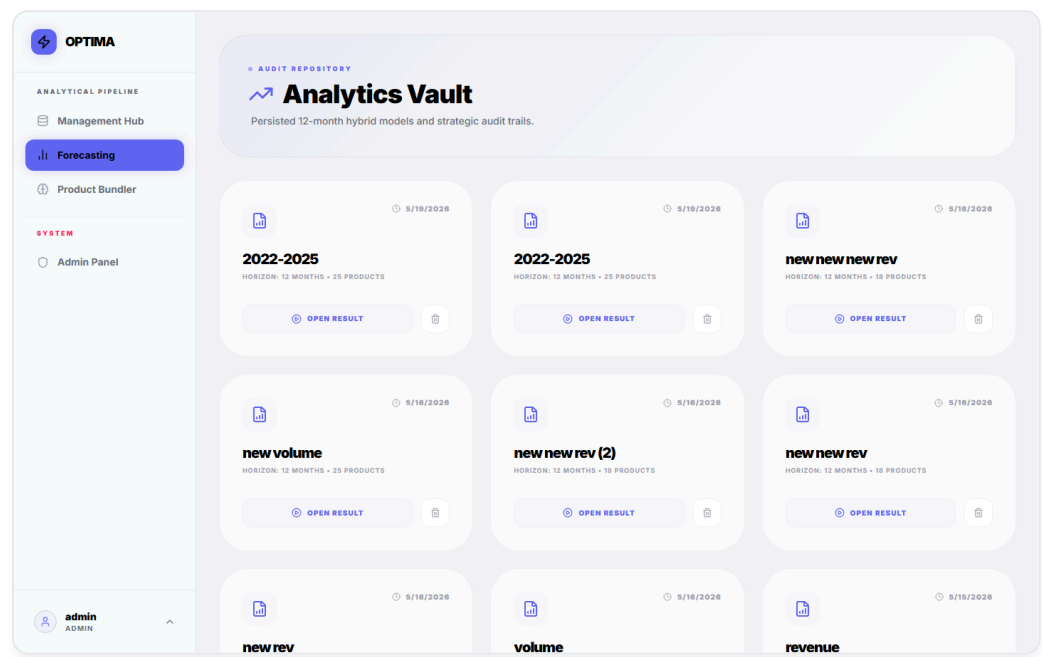
After training completes:

- If **Forecaster only** was enabled → you are redirected to the **Forecasting** page.
- If **Bundler** was enabled → you are redirected to the **Product Bundler** page with results ready for review.

7. Forecasting (Analytics Vault)

Navigate to **Forecasting** in the sidebar.

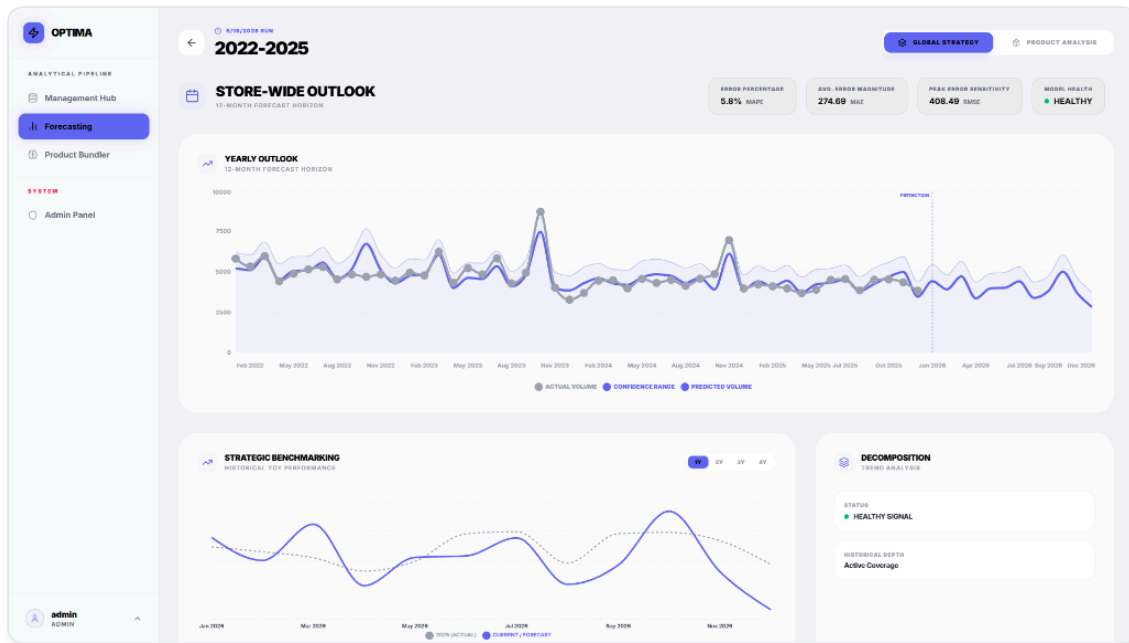
7.1 Run Browser



Displays a card grid of all saved forecast runs.

Element	Description
Run card	Shows the run name, creation date, and the number of products forecasted. Click anywhere on the card to open it.
Open Result button (on card)	Opens the run detail view for that run.
Trash icon (on card)	Deletes this forecast run after a confirmation dialog.
Empty state	If no runs exist, a message prompts you to go to the Management Hub to train your first model.

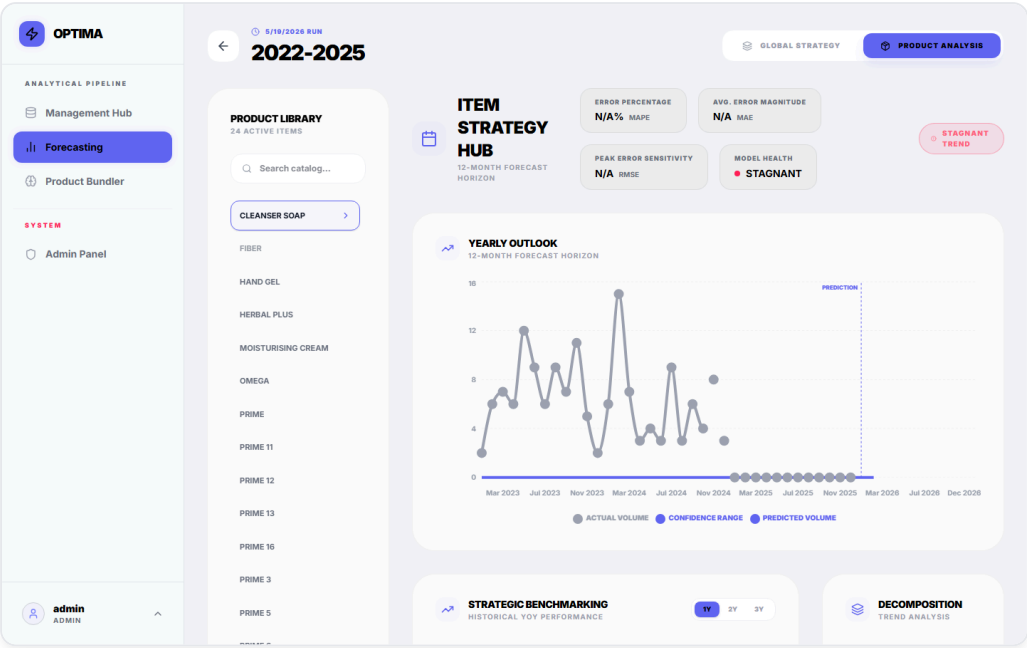
7.2 Run Detail - Global Strategy



After clicking a run card, the detail view opens.

Element	Description
Back arrow button	Returns to the Run Browser.
Global Strategy tab	Shows aggregate store-wide forecast data.
Product Analysis tab	Shows per-product forecast data with a product selector sidebar.
Yearly Outlook chart	A composed line + area chart showing actual historical sales (solid line) and 12-month forecast (accent line). Shaded region = confidence interval. A vertical "PREDICTION" line marks where the forecast begins.
Strategic Benchmarking chart	Year-over-year comparison chart showing the forecasted period vs. a prior year.
Year selector (1Y, 2Y, 3Y...)	Switches the benchmark baseline year.
Delta Variance panel	Shows the percentage and unit difference between forecast and historical benchmark. Green = exceeding baseline; Red = underperforming.
Decomposition card	Shows whether the model signal is Healthy (green dot) or Stagnant (red dot).

7.3 Run Detail - Product Analysis

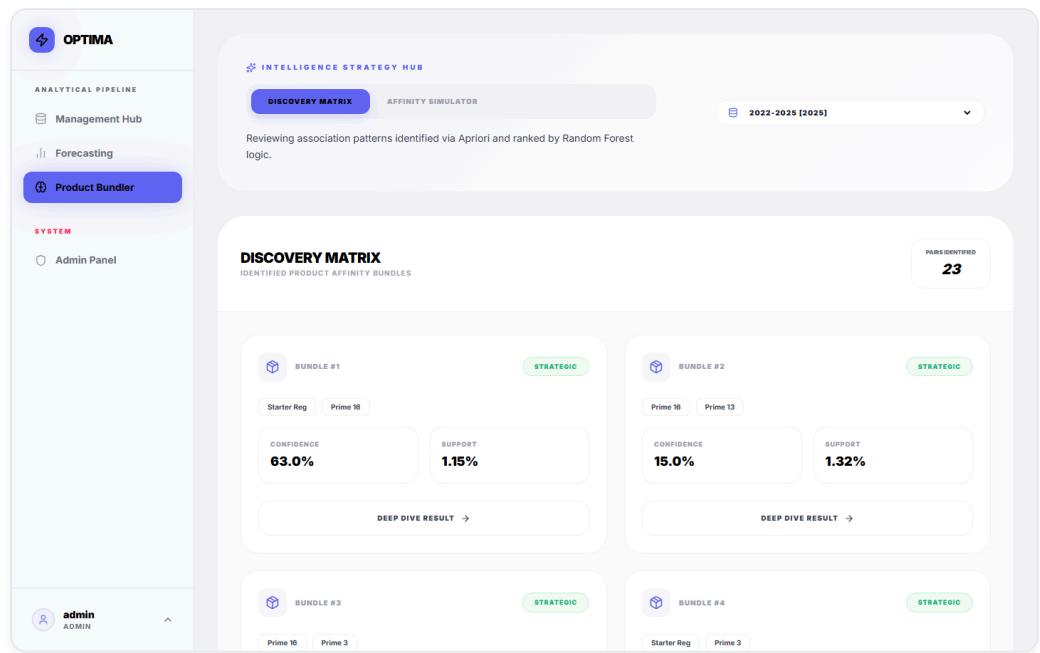


Element	Description
Product Library sidebar	Lists all products in this run.
Search catalog input	Filters the product list as you type.
Product item button	Click any product to load its individual charts.
Model Health badges	MAPE (error %), MAE (avg error magnitude), RMSE (peak error sensitivity), and a Health indicator (HEALTHY = green dot, STAGNANT = red dot).
Strategic STL Hub (if available)	Four mini-charts: Observed , Trend , Seasonal , and Remainder components of the time series.
Strategy tags	Pill badges (e.g., "SEASONAL", "TRENDING") derived from the model's analysis.

8. Product Bundler

Navigate to **Product Bundler** in the sidebar.

8.1 Discovery Matrix

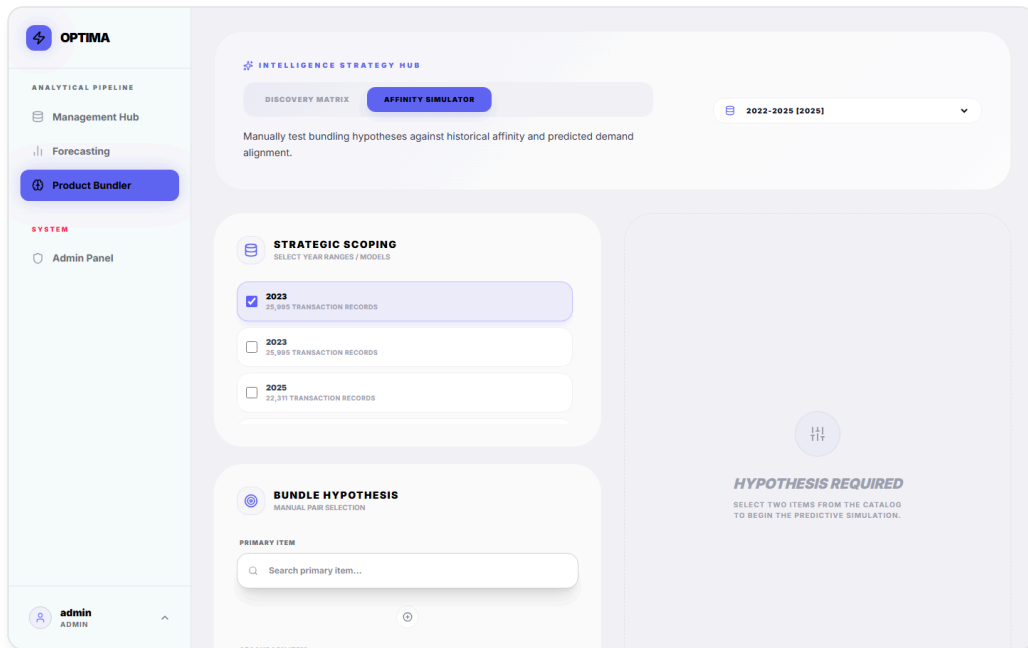


Element	Description
Discovery Matrix / Affinity Simulator tabs	Switches between modes. Active tab is highlighted in the accent color.
Run selector dropdown	Selects which saved bundler run to display.
Commit Strategy button (Sandbox mode only)	Saves unsaved sandbox bundle results permanently to the Strategy Vault.

Bundle Card

Element	Description
Bundle #N label	Sequential number for this bundle.
Badge	STRATEGIC (green) - reliable pairing. EMERGING (indigo) - growing pattern. SEASONAL (amber) - time-specific. RISK (red) - slow-mover pairing.
Item tags	Pills showing the products in this bundle.
Confidence %	How often Item B is purchased given Item A was also purchased.
Support %	How often this pair appears together across all transactions.
Deep Dive Result → button	Opens the Bundle Detail page for in-depth analysis.

8.2 Affinity Simulator



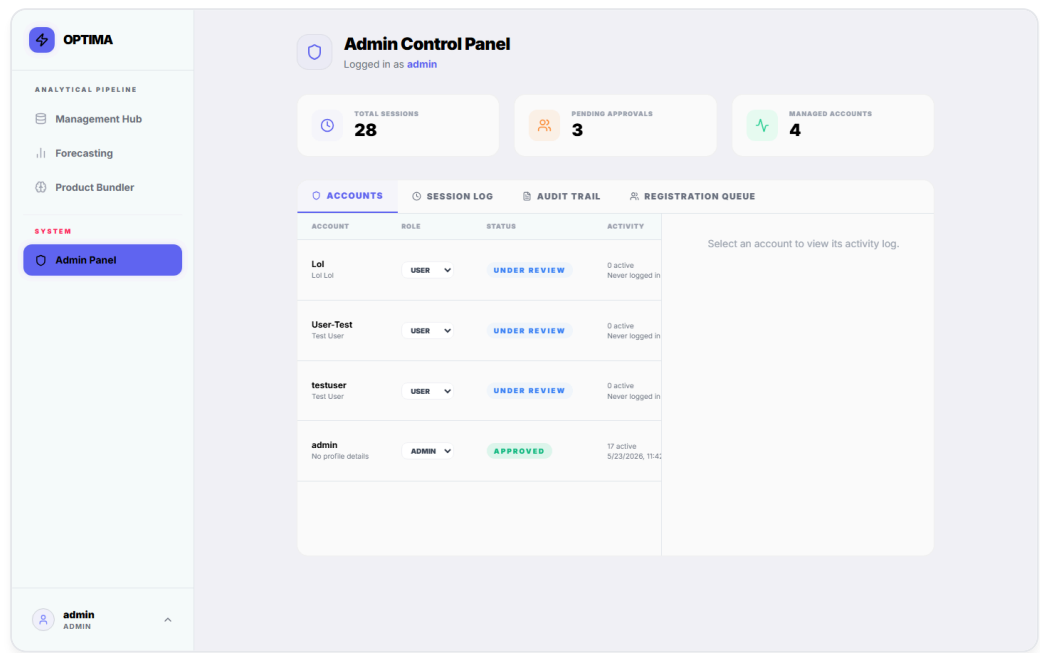
Manually test a custom product pair to get an instant bundling assessment.

Element	Description
Strategic Scoping panel	Select which datasets to include in the simulation. The active dataset is pre-selected.
Primary Item search input	Type to search and select the first product. A dropdown appears after typing one character.
X clear button (inside input)	Clears the currently selected primary item.
Secondary Item search input	Type to search and select the second product. The primary item is excluded automatically.
Simulation result panel	Appears automatically when both items are selected. Shows Badge, Model Score %, Lift, Confidence, and Strategic Rationale.
Add Simulation to Strategy Vault button (green)	Permanently saves this manual simulation result.

9. Admin Panel (Admins Only)

Navigate to **Admin Panel** in the sidebar. Only visible to users with the **ADMIN** role.

9.1 Accounts Tab

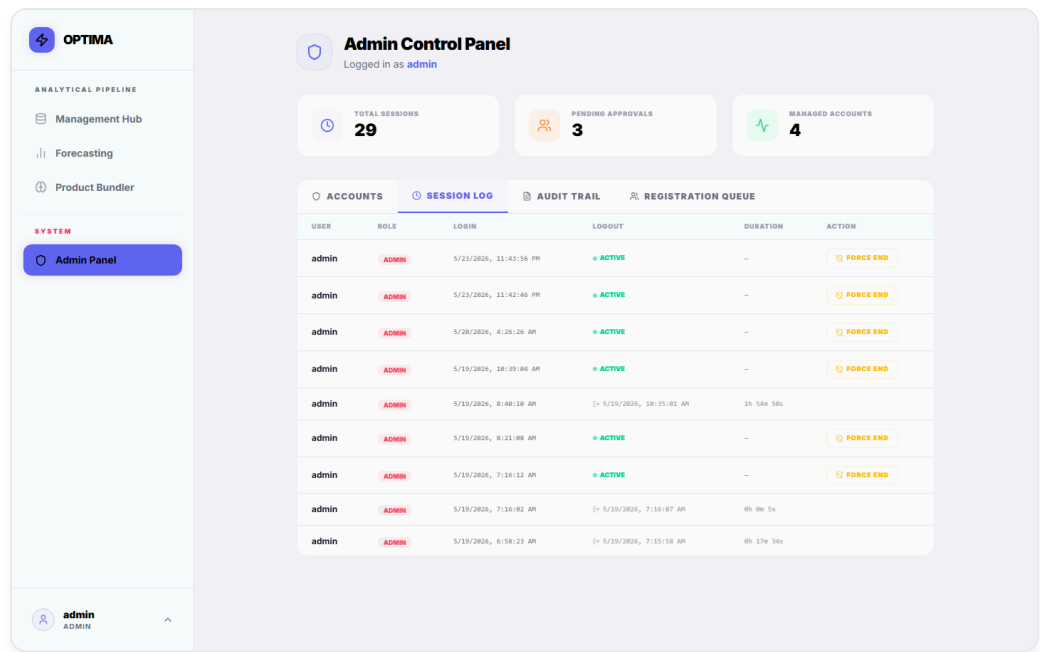


Three KPI cards at the top show: **Total Sessions**, **Pending Approvals**, and **Managed Accounts**.

Account Table Columns

Column	Description
Account	Username and display name/email. Click the username to load that user's activity log in the right panel.
Role dropdown	Instantly change a user's role between USER and ADMIN .
Status badge	APPROVED (green), PENDING (blue), BANNED (red), DENIED (red).
Activity	Number of active sessions and last login timestamp.
Controls	Ban/Reactivate and View Log buttons.

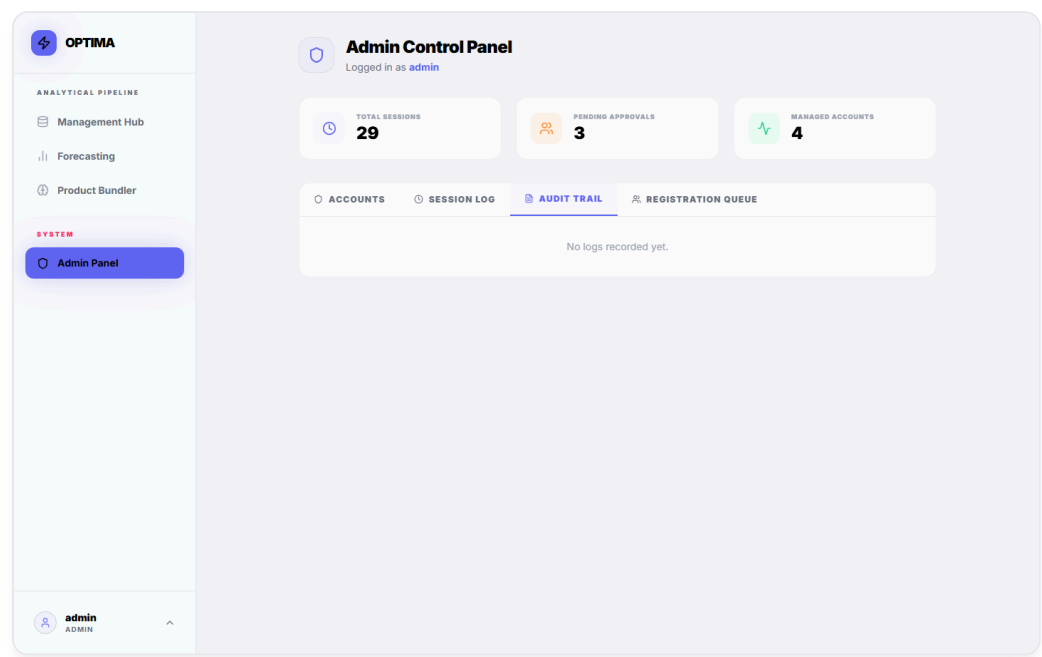
9.2 Session Log Tab



A real-time system-wide table of all login sessions.

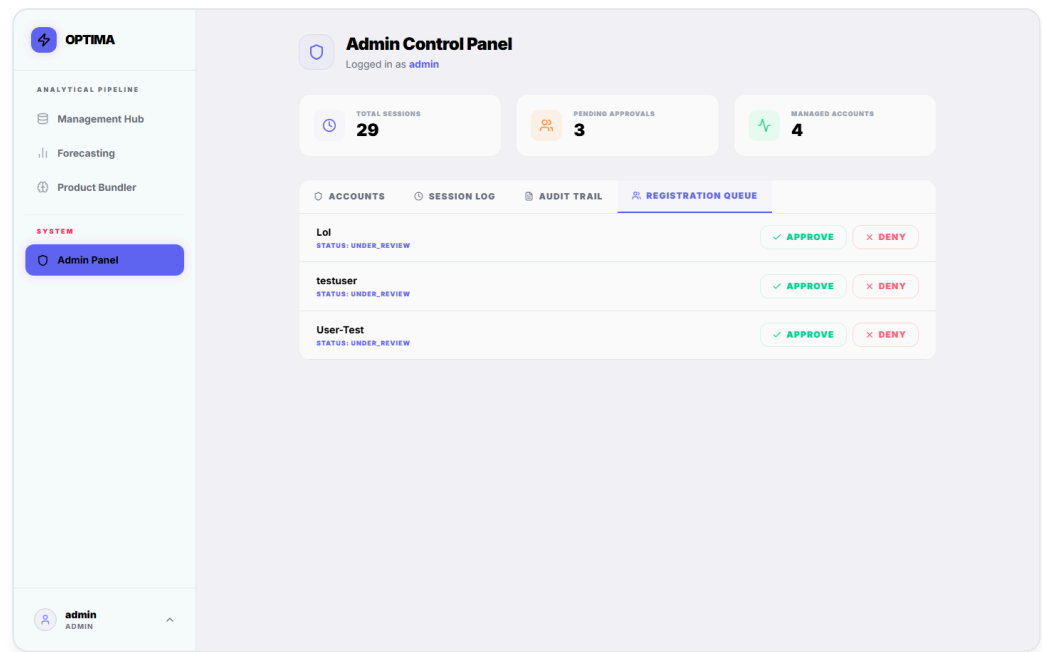
Column	Description
User	Username who initiated the session.
Role	Role used (USER or ADMIN).
Login / Logout	Timestamps, or a green pulsing Active badge if still open.
Duration	How long the session lasted.
Force End button (amber)	Immediately terminates that user's active session. A confirmation dialog appears first.

9.3 Audit Trail Tab



A chronological log of all system actions across all users - logins, role changes, approvals, and more.

9.4 Registration Queue Tab



Lists users who have registered but whose accounts have not yet been approved or denied.

Button	Description
Approve (green)	Approves the account. The user can now log in.
Deny (red)	Denies the account. The user's status becomes denied and they cannot log in.

10. Tips & Troubleshooting

Workflow: First-Time Setup (End-to-End)

1. Admin approves your account (Registration Queue)
 2. Log in → Management Hub
 3. Upload your sales CSV/Excel file (Data Ingestion)
 4. Configure item rules (mark bundles or exclude non-products)
 5. Commit the dataset
 6. Switch to Model Training tab
 7. Select the dataset, enable Forecaster and/or Bundler
 8. Name the run → Launch Training
 9. View results in Forecasting or Product Bundler
-

Password Requirements

Your password must include **all** of the following:

- Minimum 8 characters
 - At least one **uppercase** letter (A–Z)
 - At least one **lowercase** letter (a–z)
 - At least one **number** (0–9)
 - At least one **special character** (!@#\$%^&*()_+--[]{};':"\|, . < > / ?)
-

Bundle Badges Explained

Badge	Meaning	Recommended Action
STRATEGIC	High confidence + high support. A reliable pairing.	Feature prominently in promotions and store layouts.
EMERGING	Moderate confidence, growing pattern.	Monitor closely; consider a trial promotion.
SEASONAL	Strength fluctuates by time of year.	Plan seasonal bundles around peak periods.
RISK	Low confidence or declining trend.	Use for clearance promotions to move slow stock.

Model Health Indicators

Indicator	Meaning
HEALTHY (Green)	The model found meaningful trend and seasonal signals. Forecasts are reliable.
STAGNANT (Red)	The product has little to no meaningful trend variation. Consider whether this product has enough transaction history.

Common Issues

Problem	Solution
"Upload failed" error	Ensure your file is .csv , .xlsx , or .xls and contains a date column, an item/product column, and a quantity column.
Training takes very long	Normal for large datasets (>100k rows). Wait up to 30 minutes.
"Connection to server was lost mid-run"	The backend may still be processing. Wait 2–3 minutes, then check the Forecasting or Product Bundler pages for results before retrying.
No bundles appear after training	Lower the Min. Support percentage and retrain, or check if your dataset has sufficient transaction volume.
Admin Panel not visible	You must be logged in with an ADMIN role. Contact your system administrator.
Dataset not showing in Model Training	Only MASTER datasets appear. Re-upload or combine yearly datasets into a Master first.

Optima v1.0 - Smart Merchandising Platform